

Haifa K – Interpretation, Construction, and Use in the Thesis Econometrics

Draft working report for Econ H191 (Haifa SIPG K series & competition reform as a shock)

1. Purpose and scope

This report sets out a detailed conceptual and practical framework for how to **interpret, build, and use** capital (K) series for the **Haifa port system** in the Econ H191 thesis. It is written to:

1. Clarify **what “K” means** in this project and how it relates to the different institutional actors at Haifa (IPC, HPC, SIPG Bayport).
2. Explain **how SIPG/Bayport K can be present before Bayport opens**, without undermining the econometric interpretation of **Bayport’s entry as a competition-inducing “shock”**.
3. Provide clear guidance on how to **construct K series** that distinguish between:
4. Physical/financial capital installed or under construction; and
5. Capital that is actually **in use** in the production function for container throughput.
6. Spell out **how each K series is used in the event-study and mediation econometrics** (Models 1A, 1B, and 2 in the thesis).

The focus is Haifa, but the logic extends to Ashdod if we later construct similar K series for HCT (TIL) and APC.

2. Institutional and econometric background

2.1 Haifa port actors and reforms

The Haifa port complex involves three distinct institutional actors:

1. **Israel Ports Company (IPC)** – landlord authority and infrastructure owner
2. Owns the breakwaters, quays, basic marine infrastructure, and related land.
3. Invests in civil works, dredging, and base infrastructure for both Haifa-Legacy and Haifa-Bayport.
4. **Haifa Port Company (HPC)** – legacy terminal operator
5. Historically the state-owned operator of the Haifa container terminals (legacy facilities).
6. Privatized in early 2023 (sale to Adani-Gadot consortium).
7. **SIPG / Haifa Bayport Terminal Co. Ltd. (SIPG)** – new entrant operator

8. Shanghai International Port Group (SIPG) won a concession to build and operate a new deep-water container terminal, **Haifa Bayport**.
9. The Haifa Bayport SPV is a consolidated subsidiary of SIPG, with functional currency in NIS.

Key **reform events** for Haifa:

- **Competition reform (Bayport entry)** – Bayport begins commercial operations around **September 2021**. This is the date when Haifa moves from a single operator (HPC) to two operators (HPC + SIPG), and ships can meaningfully choose between terminals.
- **Privatization reform (HPC sale)** – HPC is sold and becomes a private operator around **January 2023**. This changes HPC governance and incentives but does not directly alter the physical K stock.

The thesis uses **event-study econometrics** around these reform dates:

- Model 1A: Reform $\rightarrow \ln(LP)$
- Model 1B: Reform $\rightarrow \ln(K/L)$
- Model 2: $\ln(LP) \rightarrow \ln(K/L)$ (elasticity, mediation logic)

The core mediation question is: *To what extent do reform-induced LP gains operate through capital deepening (K/L), versus other channels?*

2.2 Why Haifa K has three components

Under the landlord port model, the total capital at Haifa is naturally decomposed by **owner**:

- **K_IPC** – infrastructure capital (breakwaters, quay walls, dredging, basic access) on IPC's books.
- **K_HPC** – legacy terminal capital (cranes, yard equipment, buildings, IT) on HPC's books.
- **K_SIPG** – Bayport terminal capital (cranes, yard equipment, buildings, IT) on SIPG's books.

We therefore define:

$$K_{\text{Haifa, total}}(t) = K_{\text{IPC}}(t) + K_{\text{HPC}}(t) + K_{\text{SIPG}}(t)$$

This is the **physical/economic capital used by any terminal at the Haifa port complex** at time t . Each shekel (or RMB) of physical capital is counted once, in exactly one owner's K series; we avoid double counting by excluding lease/intangible rights when those merely duplicate underlying assets.

This owner-decomposition is the backbone for both descriptive analysis and the construction of **K/L** for the Haifa port system and its terminals.

3. Three concepts of capital: accounting, physical, and productive

To keep the econometric interpretation clean, it is crucial to distinguish three related but different notions of K:

1. Accounting capital (book K)

2. What appears on financial statements:

- Property, plant, and equipment (PPE),
- Construction in progress (CIP),
- Long-term prepayments for construction projects,
- Lease right-of-use (ROU) assets, concession intangible assets, etc.

3. This is governed by standards (IFRS, PRC GAAP, Israeli GAAP) and subject to revaluations, FX translation, and classification rules.

4. Physical capital (installed or under construction)

5. The actual stock of physical assets: breakwaters, quay walls, berths, cranes, yard equipment, tractors, buildings, IT systems, etc.

6. Includes assets that are

- fully operational, and
- still under construction but physically present (e.g., a quay still being built, cranes installed but not yet in regular service).

7. Productive (in-use) capital services

8. The subset of physical capital that is **actually used to produce container throughput** in the period —i.e., the capital that appears in the **production function** for TEU or tons.

9. Excludes purely under-construction assets that are not yet part of regular operations.

In the context of SIPG/Bayport:

- From an **accounting** and **physical** perspective, Bayport K starts to appear in SIPG's books and on the ground **from around 2018**, as we see large construction-in-progress (CIP) lines and project budgets.
- From a **productive** perspective, Bayport K almost entirely **does not contribute to Haifa's container throughput** until the terminal opens to commercial traffic ($\approx$ Sept 2021). Before that, there may be testing and training, but no meaningful container production.

This distinction allows us to: - Track the **build-up of Bayport K** during the construction phase (useful descriptively), while - Still treating **Bayport's entry as a sharp change in productive K and competition** in the econometric design.

4. Interpreting SIPG/Bayport K over time

4.1 Timeline and phases

The SIPG/Bayport project can be divided into three broad phases:

1. **Pre-project / planning phase (pre-2018)**
2. Concession awarded to SIPG; project planned and financed.

3. No significant physical Bayport K installed at Haifa yet.
4. From the thesis perspective: $K_{SIPG}(t) = 0$ and Bayport does not exist as a productive unit.
5. **Construction phase (roughly 2018–mid-2021)**
 6. IPC invests heavily in the Bayport breakwater, quay walls, dredging, and basic infrastructure.
 7. SIPG invests in cranes, yard equipment, buildings, and IT systems.
 8. SIPG's financial statements show:
 - **Long-term prepayments** for the "Israel Haifa Bayport port engineering project" (2018–2019), and
 - **Construction in progress (CIP)** entries for the Haifa Bayport project with a substantial project budget (2020–2021 and into 2022).
 9. Physically, cranes and infra are being installed; however, Bayport is still not open for regular container operations.
 10. From the productive K perspective: almost all of this capital is **under construction**, not yet delivering steady capital services.
11. **Operating phase (from September 2021 onward)**
 12. Bayport opens and starts handling container ships.
 13. A large portion of CIP is transferred to PPE (fixed assets ready for use).
 14. Bayport's cranes, yard, and equipment begin contributing to Haifa's throughput.
 15. From this point forward, **K_{SIPG} contributes to K/L and LP at Haifa.**

This timing is crucial for interpreting the **competition reform as a shock**:

- **Reform / competition "shock"** = the date when Bayport becomes an active competitor—i.e. when productive K_{SIPG} becomes operational and ships can choose Bayport vs Haifa-Legacy.
- **Pre-2021 Bayport capex** = capital under construction, economically predetermined but not yet part of the production function.

4.2 Two layers of SIPG K series

Given the above, we will maintain two related SIPG K series:

1. **$K_{SIPG,project}(t)$** – project-level capital (accounting/physical view)
2. Built directly from SIPG's financial reports (prepayments, CIP, transfers to PPE).
3. Starts to rise in 2018 as long-term prepayments and CIP accumulate.
4. Tracks the total (real) resources SIPG has committed to the Haifa Bayport project, including under-construction assets.
5. **$K_{SIPG,in_use}(t)$** – productive capital services (econometric K for K/L)
6. The portion of Bayport K that is realistically available for container handling in period t .

7. Approximately zero until Bayport begins operations (practically: until ~Sept 2021, possibly with a short ramp-up window).
8. After opening, rises as remaining CIP is transferred to PPE and additional investments are made.

These two series are conceptually linked but serve different purposes:

- **K_SIPG,project** is appropriate for descriptive figures about “capital deepening at Haifa” and for showing that the reforms involved massive new capital investment even before competition began.
- **K_SIPG,in_use** is the correct notion for **K/L in the production function** and thus for the econometric models (especially the mediation analysis).

5. Building each component of Haifa K

In practice, the Haifa K series will be built in three steps, one for each owner, plus a step to define K_in_use.

5.1 IPC K (K_IPC)

Goal: A monthly real K_IPC(t) series for Haifa-related infrastructure.

Method (brief, as this is largely settled):

1. Extract IPC's PPE note and related tables:
2. Identify Haifa-relevant assets (marine infra, dredging, berths, etc.), ideally by project or asset class.
3. Build a **PPE flows table** (opening cost, additions, transfers, disposals, closing cost; accumulated depreciation and depreciation expense).
4. Construct an annual IPC-Haifa K series using **Perpetual Inventory Method (PIM)**:
5. Choose asset lives and depreciation rates consistent with OECD Measuring Capital guidelines and with the treatment of HPC K.
6. Deflate nominal PPE values with appropriate capital goods deflators.
7. Ensure accounting consistency (stock-flow identities approximately hold).
8. Interpolate the annual series to **monthly K_IPC(t)**:
9. Typically via piecewise-linear interpolation, treating net additions as mid-year.

Interpretation:

- K_IPC represents the **shared infrastructure platform** that enables both HPC and SIPG operations.
- It is always “productive” in the sense that once built, it enables port operations; we do not distinguish under-construction vs in-use for IPC in as fine detail as for SIPG, but conceptually the same logic applies.

5.2 HPC K (K_HPC)

Goal: A monthly real K_HPC(t) series for Haifa-Legacy (HPC).

Method (already designed and largely implemented):

1. Extract HPC's Note 8/10 PPE tables by asset class:
2. Cost opening and closing, additions, transfers, capitalized interest, disposals.
3. Accumulated depreciation opening and closing, depreciation expense.
4. Build annual **PPE stock and flow tables** (Table 1 and Table 2 in the existing K pipeline).
5. Apply PIM to obtain annual real K_HPC.
6. Interpolate to monthly K_HPC(t).

Interpretation:

- K_HPC is the capital stock of the legacy operator (cranes, yard, buildings, etc.).
- It is in use throughout the period, but its level and growth may respond to both competition (Bayport entry) and privatization.

5.3 SIPG/Bayport K – project level (K_SIPG,project)

Goal: A monthly real K_SIPG,project(t) series that captures **all SIPG investment in the Haifa Bayport project** (physical/accounting view).

Data sources: SIPG consolidated financial statements (2018–2022+):

- 2018–2019: long-term prepayments for the “Israel Haifa Bayport port engineering project.”
- 2020–2021–2022 H1: detailed **CIP tables** for the “Haifa Bayport port engineering project” with project budget, opening/closing CIP, additions, and transfers to fixed assets.
- 2022+: VAT refund items and other project-related assets.

Construction steps:

1. **Raw extraction table**
2. For each report-year and note, extract project-specific values into a raw table with columns:
 - `source_id, year, page, note_id, variable_name, value, currency, units, notes`.
3. Variables include: project budget, opening CIP/prepayment, additions, transfers to PPE, other decreases.
4. **Unify currencies and deflate**

5. SIPG data appear in RMB and sometimes USD prepayments; convert to RMB then to NIS or a common real unit.
6. Deflate using appropriate capital deflators (consistent with IPC/HPC treatment).

7. **PIM on project investment path**

8. Treat **net investment I_t** as the sum of additions in each year (prepayments + CIP additions minus reversals).
9. Run PIM to obtain $K_{SIPG,project}(t)$, starting from zero prior to the project.
10. Interpolate from annual to monthly.

Interpretation:

- $K_{SIPG,project}(t)$ is the **total project capital**—it includes assets under construction and assets already in use.
- It is the right series if we want to answer: “How much capital has SIPG sunk into Haifa by a given date?”
- It is **not yet the right series for K/L** in the production function, because it counts under-construction assets that are not producing throughput.

5.4 SIPG/Bayport K – productive K (K_{SIPG,in_use})

Goal: A monthly real $K_{SIPG,in_use}(t)$ series that represents **only the SIPG capital actually in use for container handling** in period t .

This is the K that belongs in the **K/L mediator** for Models 1B and 2.

We consider two closely related options; both are acceptable if clearly documented.

Option A – Use CIP → PPE transfers and opening date

1. Use SIPG CIP tables to identify, by year, **transfers from CIP to PPE** for the Haifa Bayport project.
2. Interpret these transfers as the points when specific chunks of capital become “ready for intended use.”
3. Combine this with the known **opening date** ($\approx$ September 2021):
4. Before opening: treat transfers cautiously; either
 - delay them until opening month, or
 - include only those clearly associated with pre-opening test operations.
5. After opening: allow all transferred assets to count as in use.
6. Run PIM on these “in-use” investments I_t to obtain $K_{SIPG,in_use}(t)$, annual then monthly.

Resulting pattern:

- $K_{SIPG,in_use}(t)$ is ≈ 0 through 2018–2019.
- It begins to rise around 2020–2021, with a noticeable increase at or just before opening.
- It then continues to grow as remaining CIP is capitalized and as new equipment is added.

Pros:

- Closest to economic reality: capital becomes productive when it is installed, tested, and commissioned.
- Respects both the accounting flows and the opening date.

Cons:

- Requires judgment about exactly when pre-2021 transferred assets start contributing to regular throughput.

Option B – Stylized “all at opening” step function

For expositional clarity, especially in an honors thesis, a simpler baseline can be used:

1. Compute $K_{SIPG,project}(t)$ as in §5.3.
2. Choose a single **opening month** (e.g., 2021-09) as the point when Bayport becomes economically relevant.
3. Define $K_{SIPG,in_use}(t)$ as:
 4. 0 for all $t < \text{opening month}$;
 5. $K_{SIPG,project}(\text{opening month})$ for $t = \text{opening month}$ (large step);
 6. $K_{SIPG,project}(t)$ for $t > \text{opening month}$ (i.e., track project capital after opening).

Resulting pattern:

- $K_{SIPG,in_use}(t)$ is exactly zero until Bayport opens, then jumps to the full project capital accumulated by that date, then grows more slowly thereafter.

Pros:

- Very transparent: “the capital matters once Bayport is open.”
- Aligns tightly with the idea of **Bayport entry as a shock**.
- Easy to explain and implement.

Cons:

- Overstates the discreteness of the change; in reality, some capital is installed and potentially usable shortly before formal opening.

5.5 Choosing and documenting the K_SIPG,in_use series

Recommended approach:

- Use **Option B** as the primary specification for the thesis econometrics:
- Treat Bayport's pre-opening investment as economically relevant **only once the terminal becomes operational**.
- This matches the narrative that Bayport's **entry as a competitor** is the key shock.
- Use **Option A** as a robustness or for a more nuanced extension:
- Show that allowing a gradual ramp-up of K_SIPG,in_use(t) around 2020–2021 does not materially change the main conclusions.

In all cases, clearly write in the thesis:

"For the purpose of the K/L mediator, we define SIPG's productive capital at Haifa, K_SIPG,in_use, as the Bayport project capital that is in regular use for container operations. We treat pre-opening investment as capital under construction and count it as economically relevant only once Bayport begins commercial operations. This ensures that our K/L measure reflects productive capacity rather than construction work-in-progress."

6. Putting it together: Haifa K in the econometric models

6.1 Port-level K and K/L

For port-level analysis (Haifa as a single production unit), we define:

- **Total physical K at Haifa:**

$$K_{\text{Haifa},\text{total}}(t) = K_{\text{IPC}}(t) + K_{\text{HPC}}(t) + K_{\text{SIPG},\text{project}}(t)$$

- **Total productive K at Haifa (for K/L):**

$$K_{\text{Haifa},\text{in_use}}(t) = K_{\text{IPC}}(t) + K_{\text{HPC}}(t) + K_{\text{SIPG},\text{in_use}}(t)$$

- **Labor at Haifa:**

$$L_{\text{Haifa}}(t) = L_{\text{HPC}}(t) + L_{\text{SIPG}}(t)$$

where L_HPC and L_SIPG will eventually be built from official monthly labor-hours data.

- **Mediator:**

$$M_{\text{Haifa}}(t) = \ln(K_{\text{Haifa},\text{in_use}}(t) / L_{\text{Haifa}}(t))$$

This $M_{Haifa}(t)$ is the **K/L measure** used in:

- **Model 1B (Reform $\rightarrow$ K/L):** event-study of competition and privatization on $\ln(K/L)$.
- **Model 2 (K/L $\rightarrow$ LP):** FE regression of $\ln(LP)$ on $\ln(K/L)$ to estimate the elasticity.

The distinction between **K_Haifa,total** and **K_Haifa,in_use** allows us to:

- Describe capital deepening including the construction phase; and
- Use a K/L measure that better corresponds to the production function once Bayport opens.

6.2 Terminal-level K and K/L (competition between entrant and legacy)

When the analysis moves to **terminal-level units** (Haifa-Legacy vs Haifa-Bayport), we can define:

- **Before Bayport opening:**
 - Haifa-Legacy is the only active terminal.
 - We can assign all productive $K_{Haifa,in_use}(t)$ to Haifa-Legacy or allocate IPC K across present and future terminals by some rule.
- Haifa-Bayport terminal has $K_{SIPG,in_use_terminal}(t) = 0$ and $L_{SIPG}(t) = 0$.
- **After Bayport opening:**
 - Haifa-Legacy: $K_{HPC}(t)$ + its share of $K_{IPC}(t)$, with $L_{HPC}(t)$.
 - Haifa-Bayport: $K_{SIPG,in_use}(t)$ + its share of $K_{IPC}(t)$, with $L_{SIPG}(t)$.

The exact allocation of IPC K between terminals can follow physical rules (berth length, yard area) or a simpler equal-split assumption; the key point is that:

- **Bayport's terminal-level K series is structurally zero before opening** and then jumps/rises.
- This makes terminal-level event studies (entrant vs legacy) align with the idea of entry as a **sharp change in the set of operating K and L at the port**.

6.3 Event-study interpretation with pre-2021 Bayport construction

With the above definitions, the event-study for Haifa's competition reform looks like this:

- **Time axis:** monthly 2018–2024.
- **Treatment dummy:** $Comp_Haifa(t) = 1$ from Sept 2021 onwards (competition-on), 0 before.

For **Model 1B (Reform $\rightarrow$ K/L):**

- Pre-2018 or early years: $K_{Haifa,in_use}(t)$ reflects only IPC + HPC K.
- 2018–mid-2021: Bayport project K is being built but **$K_{SIPG,in_use}(t)$ is still zero** (under Option B) or very small; $K_{Haifa,in_use}(t)$ mostly tracks IPC + HPC.
- Sept 2021 onward: $K_{SIPG,in_use}(t)$ jumps/rises; $K_{Haifa,in_use}(t)$ and thus K/L may show a discrete and/or accelerated increase.

Interpreting the event-study coefficients:

- **Leads of treatment ($m < 0$):** tell us whether there is a visible pre-trend in K/L prior to competition when we define K/L based on **productive K**, not construction work.
- **Post-treatment bins ($m \geq 0$):** capture changes in K/L associated with Bayport's entry as a competitor, i.e., the combined effect of new productive capital and any changes in labor.

The construction-phase K (2018–2021) does **not** contaminate these estimates, because it is treated as **under construction** until Bayport opens in the K_SIPG, in_use definition.

6.4 Mediation logic

The mediation analysis connects:

1. **Competition $\rightarrow$ LP (Model 1A):** event-study on $\ln(LP)$ around Bayport entry.
2. **Competition $\rightarrow$ K/L (Model 1B):** event-study on $\ln(K/L)$ using $K_Haifa, in_use(t)$.
3. **K/L $\rightarrow$ LP (Model 2):** FE regression of $\ln(LP_{\{i,t\}})$ on $\ln(K/L_{\{i,t\}})$ to estimate the elasticity θ .

By using K_SIPG, in_use in the K/L measure, we ensure that:

- The “**mediator**” K/L reflects capital that is plausibly part of the production function in each period.
- **Pre-opening Bayport construction** affects neither K/L nor LP in a mechanical way, so the **reform can still be interpreted as a shock**.

The decomposition of the reform effect into

- Indirect (via K/L), and
- Residual (other channels)

is therefore conceptually clean: the capital that “turns on” in the K/L mediator is precisely the capital that Bayport brings into active service once it starts competing.

7. Summary and practical guidelines

1. **Separate project K from productive K:**
2. Build a full $K_SIPG, project(t)$ series from SIPG's CIP and prepayment data.
3. Derive $K_SIPG, in_use(t)$ by treating pre-opening capital as under construction and switching it into the productive stock when Bayport opens (Option B as baseline; Option A as a nuance).
4. **Define Haifa K appropriately for each use:**
5. **Descriptive total K:** use $K_Haifa, total(t) = K_IPC + K_HPC + K_SIPG, project$.
6. **Econometric K/L mediator:** use $K_Haifa, in_use(t) = K_IPC + K_HPC + K_SIPG, in_use$.
7. **Maintain K as additive by owner:**

8. Ensure no double counting by excluding lease/intangible rights that duplicate physical assets in IPC or others.

9. Map asset classes consistently across IPC, HPC, and SIPG.

10. Preserve the reform as a shock:

11. Define competition treatment based on **Bayport opening**, not on first capex.

12. Use K_{SIPG,in_use} in K/L so that pre-2021 Bayport construction does not artificially inflate pre-treatment K/L.

13. Document assumptions and run robustness checks:

14. Clearly state the treatment of pre-opening investment in the thesis text.

15. Show that alternative plausible definitions of K_{SIPG,in_use} (e.g., smoother ramp-up) do not overturn the main qualitative conclusions.

Following this framework, the Haifa K series will be both **economically interpretable** and **econometrically coherent**, allowing the thesis to use Bayport's entry as a genuine competition shock while still acknowledging the multi-year investment process behind the reform.